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Stroke: Vascular and Interventional Neurology

ORIGINAL RESEARCH

Visual Tools for Informed Decision-Making in Large-Core Thrombectomy for Acute Ischemic Stroke

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BACKGROUND: Recent randomized clinical trials have demonstrated that endovascular thrombectomy (EVT) improves outcomes in patients with large ischemic cores. Despite these findings, hyperacute decision-making in large-core acute ischemic stroke remains challenging, as patients continue to face high rates of disability and mortality. Clear communication of risks and benefits is essential, and visual aids may improve comprehension in emergency settings.

METHODS: Ninety-day modified Rankin Scale score distributions were pooled from 6 large-core thrombectomy trials: RESCUE-Japan LIMIT (Recovery by Endovascular Salvage for Cerebral Ultra-Acute Embolism-Japan Large Ischemic Core Trial), SELECT-2 (Randomized Controlled Trial to Optimize Patient's Selection for Endovascular Treatment in Acute Ischemic Stroke), ANGEL-ASPECT (Study of Endovascular Therapy in Acute Anterior Circulation Large Vessel Occlusive Patients With a Large Infarct Core), TENSION (Efficacy and Safety of Thrombectomy in Stroke With Extended Lesion and Extended Time Window), TESLA (Thrombectomy for Emergent Salvage of Large Anterior Circulation Ischemic Stroke), and LASTE (Large Stroke Therapy Evaluation). Outcomes included functional independence (modified Rankin Scale score, 0–2), acceptable outcome (modified Rankin Scale score, 0–3), symptomatic intracranial hemorrhage, and decompressive hemicraniectomy. Benefit per hundred and harm per hundred were calculated by comparing EVT with medical management. Net benefit was defined as benefit per hundred minus harm per hundred. Visual decision aids, including a Single Personograph Choice Consequence Matrix, were developed to illustrate the pooled results.

RESULTS: Across 1872 patients, EVT increased rates of functional independence (19.5% versus 7.5%) and acceptable outcomes (36.5% versus 20.0%) compared with medical management. EVT also reduced severe disability (12.2% versus 20.6%) and mortality (31.0% versus 37.2%). Rates of symptomatic intracranial hemorrhage were 5.5% in the EVT arm and 3.2% in the medical management arm, while hemicraniectomy occurred in 12.1% and 10.4%, respectively, corresponding to an excess harm of 3.9%. The calculated benefit per hundred ranged from 16.5% (modified Rankin Scale score, 0–2) to 55.6% (ordinal shift), with an average of 36.1%. The visual aid illustrated that when scaled to a cohort of 100 patients receiving EVT, 40 would be expected to benefit, 4 to experience harm, 31 to die, and 29 to show no difference compared with medical management.

CONCLUSIONS: EVT for large-core acute ischemic stroke provides substantial benefit despite increased risks of hemorrhage and surgical rescue. Visual decision aids based on pooled trial data offer an objective method for presenting outcomes, supporting informed and timely decision-making in acute stroke care.

GRAPHIC ABSTRACT: A graphic abstract is available for this article.

Key Words: clinical decision-making ■ endovascular procedures ■ infarction ■ ischemic stroke ■ thrombectomy

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Historically, patients with large-core infarcts were considered at high risk for poor outcomes despite attempted revascularization. Consequently, patients with Alberta Stroke Program Early Computed Tomography Score (ASPECTS) of <6 were excluded from the pivotal endovascular thrombectomy (EVT) trials of 2015

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Nonstandard Abbreviations and Acronyms

AIS	acute ischemic stroke
ANGEL-ASPECT	Study of Endovascular Therapy in Acute Anterior Circulation Large Vessel Occlusive Patients With a Large Infarct Core
ASPECTS	Alberta Stroke Program Early Computed Tomography Score
BPH	benefit per hundred
EVT	endovascular thrombectomy
LASTE	Large Stroke Therapy Evaluation
mRS	modified Rankin Scale
RESCUE-Japan LIMIT	Recovery by Endovascular Salvage for Cerebral Ultra-Acute Embolism-Japan Large Ischemic Core Trial
SELECT-2	Randomized Controlled Trial to Optimize Patient's Selection for Endovascular Treatment in Acute Ischemic Stroke
sICH	symptomatic intracranial hemorrhage
TENSION	Efficacy and Safety of Thrombectomy in Stroke With Extended Lesion and Extended Time Window
TESLA	Thrombectomy for Emergent Salvage of Large Anterior Circulation Ischemic Stroke

CLINICAL PERSPECTIVE

What Is New?

- This pooled analysis of 6 large-core randomized thrombectomy trials (RESCUE-Japan LIMIT [Recovery by Endovascular Salvage for Cerebral Ultra-Acute Embolism-Japan Large Ischemic Core Trial], SELECT -2 [Randomized Controlled Trial to Optimize Patient's Selection for Endovascular Treatment in Acute Ischemic Stroke], ANGEL-ASPECT [Study of Endovascular Therapy in Acute Anterior Circulation Large Vessel Occlusive Patients With a Large Infarct Core], TENSION [Efficacy and Safety of Thrombectomy in Stroke With Extended Lesion and Extended Time Window], TESLA [Thrombectomy for Emergent Salvage of Large Anterior Circulation Ischemic Stroke], and LASTE [Large Stroke Therapy Evaluation]) demonstrates that endovascular thrombectomy provides significant functional benefit even in patients with extensive infarction volumes historically considered ineligible for treatment.
- We introduce a novel Single Personograph Choice Consequence Matrix that visually depicts benefit and harm per hundred, enabling rapid comprehension of procedural trade-offs in emergency decision-making.
- Across 1872 patients, the calculated benefit per hundred was approximately 36% and the harm per hundred was 4%, yielding a substantial net clinical benefit that remains consistent across Alberta Stroke Program Early Computed Tomography Score, age, and National Institutes of Health Stroke Scale subgroups.

What Are the Clinical Implications?

- Endovascular thrombectomy (EVT) should be strongly considered in appropriately selected large-core acute ischemic stroke patients, as it meaningfully improves outcomes despite modest increases in procedural risk.
- Visual decision aids derived from pooled trial data can enhance communication, counteract negativity bias, and facilitate shared decision-making for patients and families during hyperacute stroke presentations.
- Integrating such tools into clinical workflows may improve understanding of EVT outcomes, align patient expectations, and promote equitable, evidence-based adoption of thrombectomy in large-core stroke care.

and 2018.[1–8] However, a series of recently published randomized clinical trials have demonstrated improved disability and poststroke mortality for these patients when treated with EVT. These trials include RESCUE-Japan LIMIT (Recovery by Endovascular Salvage for Cerebral Ultra-Acute Embolism-Japan Large Ischemic Core Trial),[9] SELECT-2 (Randomized Controlled Trial to Optimize Patient's Selection for Endovascular Treatment in Acute Ischemic Stroke),[10] ANGEL-ASPECT (Study of Endovascular Therapy in Acute Anterior Circulation Large Vessel Occlusive Patients With a Large Infarct Core),[11] TENSION[12] (Efficacy and Safety of Thrombectomy in Stroke With Extended Lesion and Extended Time Window), TESLA (Thrombectomy for Emergent Salvage of Large Anterior Circulation Ischemic Stroke),[13] and LASTE (Large Stroke Therapy Evaluation).[14] Clinical decision-making in patients with large-core acute ischemic stroke (AIS) can be difficult, as patients are still at risk of hemorrhagic transformation and have nonnegligible rates of

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disability and mortality due to the larger area of irreversible ischemia at the time of presentation compared with smaller core AIS.[15] Patients and their family members are required to make a life-altering decision in minutes, often with limited medical guidance or clarity. Visual aids have been shown to serve as powerful adjunctive tools in shared decision-making. They allow patients and families to better understand complex information in a short time, allowing for greater participation in decision-making and better risk perception.[16] We aimed to produce a visual aid that could quickly and effectively convey the potential benefits and risks of large-core AIS EVT in emergency settings. Similar approaches have been taken in decision-making for thrombolysis, as well as EVT for late-presenting AIS.[17,18]

METHODS

The authors declare that all supporting data are available within this article.

Benefit per Hundred Calculations

Global disability outcome distributions from the thrombectomy and control arms were pooled between the 6 randomized controlled trials. Ninety-day modified Rankin Scale (mRS) score distributions for each randomized group were weighted by sample size when pooling, yielding Figure 1. This was then used to generate 100-person icon arrays (Kuiper-Marshall personographs) from this disability distribution, yielding Figure 2.

After pooling the 6 trials, we calculated the rates of functional independence (90-day mRS score, 0–2), acceptable outcome (90-day mRS score, 0–3), and shift in mRS score. Using these values, we calculated the benefit per hundred (BPH) for all the above outcomes including binary outcomes, as well as ordinal outcomes in the direction of benefit or harm. In addition, we calculated rates of symptomatic intracranial hemorrhage (sICH) and decompressive hemicraniectomy. These 2 outcomes were combined to derive the harm per hundred. This approach was chosen to provide a conservative upper-bound estimate of treatment-related harm, ensuring that the risk of EVT was not underestimated. While some patients may experience both complications, each event was directly extracted from trial-reported rates, and the combined value represents a rigorous and intentionally conservative measure. We calculated net benefit as the difference between BPH and harm per hundred.

We generated 2 visual decision aids based on the 6-level mRS score and pooled randomized controlled trial data. We generated a Single Personograph Choice Consequence Matrix. The purpose of these figures is to illustrate the shifts in outcomes that take place because of a particular intervention. We used the calculated BPH, harm per hundred, and net benefit to generate the outcomes. Three-color shading was used to demonstrate changes in the 90-day mRS outcomes between patients treated with EVT versus those managed medically. The green color was used for patients who had less disability following the procedure. Dark green was used for patients who had a 90-day mRS score of 0 to 2, and light green was used for those who had a 90-day mRS score that had an acceptable outcome of mRS score 0 to 3. White color was used for patients who had no change in the 90-day mRS outcome. Those who suffered either sICH or underwent decompressive hemicraniectomy were demonstrated with red. A black line was used to describe death (mRS score, 6).

Subgroup Meta-Analyses

In addition to the pooled overall analysis, subgroup meta-analyses were performed across the 6 large-core randomized controlled trials. Ordinal logistic regression models were used to generate odds ratios for the distribution of 90-day mRS scores by treatment group. Subgroups examined included younger versus older patients, lower versus higher baseline National Institutes of Health Stroke Scale score, and ASPECTS of 0 to 2 versus ASPECTS of 3 to 5. Because RESCUE-Japan LIMIT reported dichotomous mRS score of 0 to 3 outcomes and TESLA used utility-weighted mRS score, trial-specific odds ratios were extracted directly from published data or approximated using exponentiation of reported coefficients. Forest plots were then generated for each subgroup comparison.

RESULTS

The global disability outcome distributions from the 6 randomized clinical trials were pooled to study the

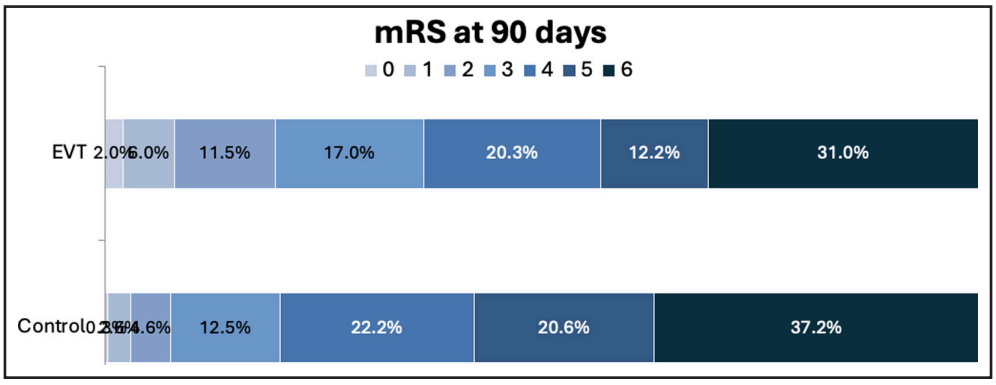

Figure 1. Ordinal shift bars demonstrating the modified Rankin Scale (mRS) score at 90 days by treatment group.
EVT indicates endovascular thrombectomy.

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Figure 2. Single personograph choice consequence matrix illustrating the outcome of large-core patients treated with mechanical thrombectomy.

effects of EVT versus medical management in large-core AIS. 0.3% of patients in the medical management arm achieved an mRS score of 0 compared with 2.0% in the EVT group. 2.6% of patients in the medical management group achieved an mRS score of 1 compared with 6.0% in the EVT group. 4.6% of patients achieved an mRS score of 2 in the medical management group compared with 11.5% in the EVT group. 12.5% of patients in the medical management arm achieved an mRS score of 3 compared with 17.0% in the EVT arm. 22.2% of patients achieved an mRS score of 4 compared with 20.3 in the EVT arm. 20.6% of patients in the medical management arm achieved an mRS score of 5 compared with 12.2% in the EVT arm. Last, 37.2% of the patients in the medical management had an mRS score of 6 at 90 days compared with 31.0% in the EVT arm (Table 1). These are also demonstrated as mRS ordinal shift bars in Figure 1.

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Table 1. Distribution of mRS Score at 90 Days by Treatment Group

mRS score	Medical management	EVT
0	0.3%	2.0%
1	2.6%	6.0%
2	4.6%	11.5%
3	12.5%	17.0%
4	22.2%	20.3%
5	20.6%	12.2%
6	37.2%	31.0%
Total	100%	100%
No. of patients	931	941

EVT indicates endovascular thrombectomy; and mRS, modified Rankin Scale.

We similarly pooled the rates of sICH in each trial. Individual rates of sICH in each trial are demonstrated in Table 2. The pooled analysis demonstrated an sICH rate of 3.2% in the medical management arm compared with 5.5% in the EVT arm. These rates were used to calculate a rate of excess treatment harm of 2.3%. A similar analysis was conducted for decompressive hemicraniectomy. These results are displayed in Table 3. The pooled rate of decompressive hemicraniectomy in the medical management arm was 10.4% compared with the rate of 12.1% in the EVT arm, yielding an excess treatment harm of 1.7%. To derive the harm per hundred, the excess rates of sICH and hemicraniectomy were combined, resulting in a conservative estimate of 3.9%. This method was chosen to provide an upper-bound assessment of treatment-related harm, recognizing the possibility of overlap between these complications, and ensures that the risk associated with EVT is not underestimated.

Table 2. Rates of sICH by Trial

Trial	EVT	Medical management
SELECT-2	1	2
ANGEL-ASPECT	14	6
RESCUE-Japan LIMIT	9	5
TESLA	6	2
TENSION	7	6
LASTE	15	9
Total	52	30
Percentage	5.5%	3.2%
Treatment excess of sICH	2.3%	

ANGEL-ASPECT indicates Study of Endovascular Therapy in Acute Anterior Circulation Large Vessel Occlusive Patients With a Large Infarct Core; EVT, endovascular thrombectomy; LASTE, Large Stroke Therapy Evaluation; RESCUE-Japan LIMIT, Recovery by Endovascular Salvage for Cerebral Ultra-Acute Embolism-Japan Large Ischemic Core Trial; SELECT-2, Randomized Controlled Trial to Optimize Patient's Selection for Endovascular Treatment in Acute Ischemic Stroke; sICH, symptomatic intracranial hemorrhage; TENSION, Efficacy and Safety of Thrombectomy in Stroke With Extended Lesion and Extended Time Window; and TESLA, Thrombectomy for Emergent Salvage of Large Anterior Circulation Ischemic Stroke.

Table 3. Rates of Hemicraniectomy by Trial

Trial	EVT	Medical management
SELECT-2	29	26
ANGEL-ASPECT	10	14
RESCUE-Japan LIMIT	17	8
TESLA	33	22
TENSION	11	9
LASTE	14	19
Total	114	98
Percentage	12.1%	10.4%
Treatment excess of hemicraniectomy	1.6%	

ANGEL-ASPECT indicates Study of Endovascular Therapy in Acute Anterior Circulation Large Vessel Occlusive Patients With a Large Infarct Core; EVT, endovascular thrombectomy; LASTE, Large Stroke Therapy Evaluation; RESCUE-Japan LIMIT, Recovery by Endovascular Salvage for Cerebral Ultra-Acute Embolism-Japan Large Ischemic Core Trial; SELECT-2, Randomized Controlled Trial to Optimize Patient's Selection for Endovascular Treatment in Acute Ischemic Stroke; TENSION, Efficacy and Safety of Thrombectomy in Stroke With Extended Lesion and Extended Time Window; and TESLA, Thrombectomy for Emergent Salvage of Large Anterior Circulation Ischemic Stroke.

The net BPH was calculated by summing the BPH by calculating the excess benefit associated with EVT for each mRS score range and summing these values after which we subtracted the net harm, which was ≈4% (combined sICH and hemicraniectomy excess treatment harm rates). Using all dichotomous splits (0–1 through 0–5/6) led to the calculation of a maximum BPH of 55.6%, whereas using only a dichotomous split of 0 to 2 led to a minimum BPH of 16.5%. We used the average of these 2 BPH, which was 36.1%. We then used these numbers in Figure 2 to portray 40 of 100 patients benefited from thrombectomy, with 12 achieving an mRS score of 0 to 2 (dark green) and 28 with positive mRS shift although not achieving an mRS score of 0 to 2 (light green). Four patients were shown as red as these experienced excess treatment harm. Thirty-one patients were shown with a black line underneath to demonstrate an mRS score of 6. Last, the remaining 29 patients had no change and were displayed as white. Patients with an mRS score of 6 (death) were specifically denoted by black lines; in this framework, death was considered neutral only when the probability of death was identical in both treatment groups, reflecting the absence of a treatment-related shift rather than implying neutrality of death as a clinical outcome.

Subgroup Analyses

ASPECTS

Subgroup analysis by ASPECTS demonstrated a consistent benefit of EVT across core size categories. In patients with ASPECTS of 0 to 2, EVT was associated with improved 90-day outcomes compared with medical management. Similarly, in patients with ASPECTS

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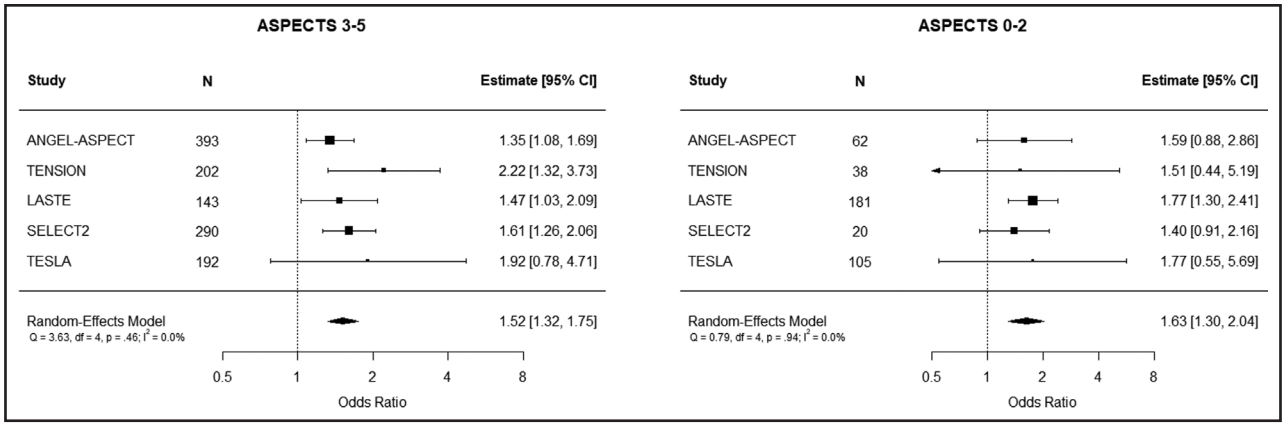

Figure 3. Forest plot of initial Alberta Stroke Program Early Computed Tomography Score (ASPECTS) and 90-day outcomes in large-core thrombectomy trials.
ANGEL-ASPECT indicates Study of Endovascular Therapy in Acute Anterior Circulation Large Vessel Occlusive Patients With a Large Infarct Core; LASTE, Large Stroke Therapy Evaluation; SELECT-2, Randomized Controlled Trial to Optimize Patient's Selection for Endovascular Treatment in Acute Ischemic Stroke; TENSION, Efficacy and Safety of Thrombectomy in Stroke With Extended Lesion and Extended Time Window; and TESLA, Thrombectomy for Emergent Salvage of Large Anterior Circulation Ischemic Stroke.

of 3 to 5, EVT also conferred benefit. RESCUE-Japan LIMIT was excluded from this analysis because patients with only ASPECTS of 3 to 5 were enrolled. The direction and magnitude of effect were consistent across both ASPECTS subgroups (Figure 3).

Age

Subgroup analysis stratified by age demonstrated EVT benefit across both younger and older patients. Odds ratios favored EVT irrespective of age group, indicating that treatment benefit is not confined to younger populations (Figure 4).

National Institutes of Health Stroke Scale

Subgroup analysis by baseline National Institutes of Health Stroke Scale score also showed consistent EVT benefit across lower and higher severity strokes. Patients with both less severe and more severe baseline deficits derived improved outcomes with EVT compared with medical management (Figure 5).

DISCUSSION

In our study, we created visual tools to assist in informed decision-making in EVT for large-core AIS. These visual aids, specifically the Single Personograph Choice Consequence Matrix, were created to clearly illustrate the potential benefits and risks associated with EVT, making complex clinical data more accessible and understandable for both patients and providers.

The pooled data from the 6 recent groundbreaking randomized clinical trials once again demonstrate the benefit of EVT for large-core AIS. The mRS distributions

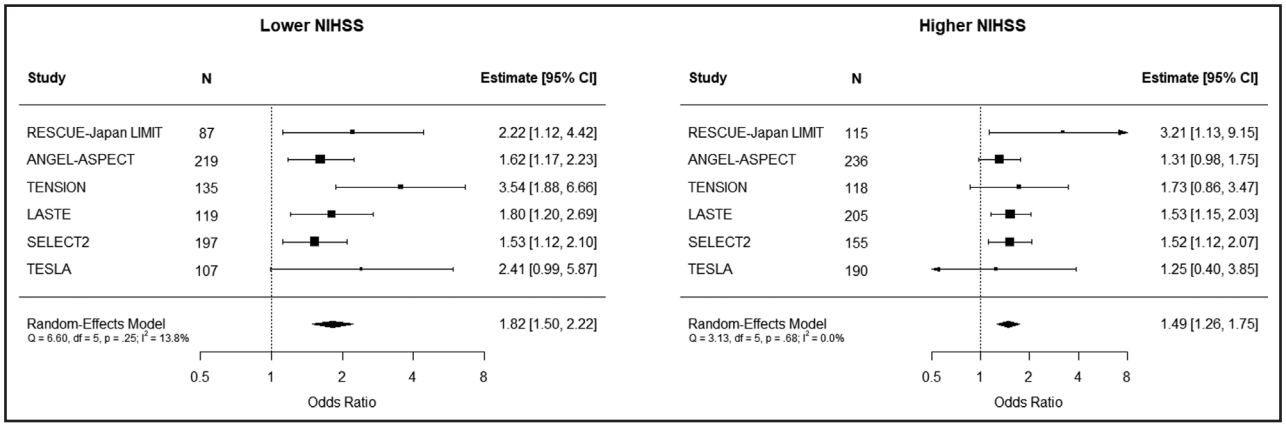

Figure 4. Forest plot of National Institutes of Health Stroke Scale (NIHSS) outcomes in large-core thrombectomy trials.
ANGEL-ASPECT indicates Study of Endovascular Therapy in Acute Anterior Circulation Large Vessel Occlusive Patients With a Large Infarct Core; LASTE, Large Stroke Therapy Evaluation; RESCUE-Japan LIMIT, Recovery by Endovascular Salvage for Cerebral Ultra-Acute Embolism-Japan Large Ischemic Core Trial; SELECT-2, Randomized Controlled Trial to Optimize Patient's Selection for Endovascular Treatment in Acute Ischemic Stroke; TENSION, Efficacy and Safety of Thrombectomy in Stroke With Extended Lesion and Extended Time Window; and TESLA, Thrombectomy for Emergent Salvage of Large Anterior Circulation Ischemic Stroke.

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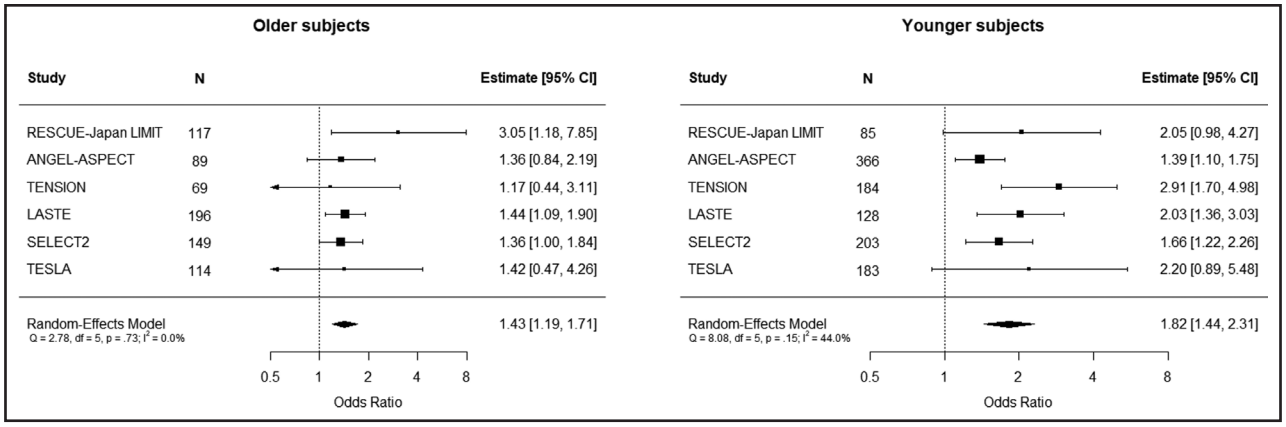

Figure 5. Forest plot of age and 90-day outcomes in large-core thrombectomy trials.
ANGEL-ASPECT indicates Study of Endovascular Therapy in Acute Anterior Circulation Large Vessel Occlusive Patients With a Large Infarct Core; LASTE, Large Stroke Therapy Evaluation; RESCUE-Japan LIMIT, Recovery by Endovascular Salvage for Cerebral Ultra-Acute Embolism-Japan Large Ischemic Core Trial; SELECT-2, Randomized Controlled Trial to Optimize Patient's Selection for Endovascular Treatment in Acute Ischemic Stroke; TENSION, Efficacy and Safety of Thrombectomy in Stroke With Extended Lesion and Extended Time Window; and TESLA, Thrombectomy for Emergent Salvage of Large Anterior Circulation Ischemic Stroke.

of patients treated with EVT compared with medical management had strikingly better outcomes. The ordinal 90-day mRS analysis demonstrates higher rates of each mRS score level from 0 to 3 and lower rates of each ordinal level from 4 to 6. When analyzed in a dichotomous fashion, this benefit was reproduced. This study aimed to weigh this benefit against the excess treatment harm and then to demonstrate that in an easily appreciable manner with the visual representation. The excess treatment harm was estimated at 4%. Using that estimate, we demonstrated a 36% BPH for EVT compared with medical management. The BPH of EVT surpasses the BPH of most cardiac interventions such as percutaneous coronary interventions,[16] coronary artery bypass graft,[19] or pacemaker placement.[20] For reference, the BPH of percutaneous coronary intervention compared with fibrinolytic therapy is 2% in patients with an ST-segment–elevation myocardial infarction.[16] This analysis and the development of visual aid tools are imperative as indications for EVT continue to expand, and decision-making becomes more complex. Unlike decision-making in earlier eras of thrombectomy, where patients were often choosing between disability and clear functional benefit, large-core EVT presents a more nuanced challenge. Large-core AIS has a morbid prognosis, and the rates of mortality in our pooled analysis were 37.2% in the medical management arm and 31.0% in the EVT arm. Despite this grave prognosis, EVT can still result in a substantial improvement in patient outcomes. In this pooled analysis, nearly 20% of patients achieved functional independence (mRS score, 0–2) with EVT, and when including the mRS score of 3, this percentage increased to 37%.

The subgroup analyses further support the robustness of the EVT benefit in large-core stroke. Across key baseline characteristics, including age, National Institutes of Health Stroke Scale severity, and ASPECTS category, EVT was consistently associated with improved 90-day outcomes compared with medical management. These findings strengthen the validity of our visual decision aids, as they suggest that the pooled estimates are not driven by a single subgroup but rather reflect a broad and consistent treatment effect across clinically relevant strata. Although some trials reported outcomes using slightly different scales (dichotomous mRS or utility-weighted mRS), sensitivity analyses demonstrated that exclusion of these trials did not materially change the conclusions. It is important to note that our study was not intended to serve as a systematic review or meta-analysis, as several such analyses have already been conducted on this topic. Rather, the purpose of our study was to review 90-day functional outcomes from large-core stroke trials and create a practical visual aid for the health care setting.

Using a visual aid such as the one created in this study can help health care providers discuss the nuances of large-core EVT with patients and their families. By visually illustrating the distribution of potential outcomes, including the probabilities of significant improvement and the risks of adverse events, providers can more effectively communicate the complex tradeoffs involved in the decision to pursue EVT. In theory, this can lead to better-aligned expectations and more personalized patient care although it has not formally been tested in the clinical setting. There are ethical implications with the use of visual aids in emergent decision-making; previous studies have shown that visual aids increase confidence and comfort in shared decision-making, especially in low health literacy and vulnerable populations.[21] An important limitation is that this study is based on 90-day outcomes, which remain the standard primary end point in acute stroke trials but do not capture long-term cognitive and functional recovery. These domains are not well

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characterized in the existing large-core randomized controlled trials although post hoc and longitudinal data are beginning to emerge and will be important to integrate into future decision-making tools. Another limitation to mention is that the visual figures presented in this study were generated using pooled data from randomized clinical trials. The outcomes depicted may not be fully generalizable to patients who do not meet the original trial inclusion criteria.

In addition, owing to negativity bias,[22] it is common for both patients and providers to weigh the risks of the procedure more heavily than the benefits. Using a visual aid such as the one created in this study may help combat this bias by visually contextualizing the risk. This clear visual representation helps bridge the gap between clinical evidence and patient comprehension, ensuring that the decision to proceed with EVT is made with a comprehensive understanding of the potential outcomes.

CONCLUSIONS

In conclusion, EVT for large-core AIS presents a significant therapeutic opportunity to reduce disability and improve acceptable outcomes, despite the notable risks. The visual decision aids developed in this study provide a crucial resource for clinicians and patients, facilitating informed, timely, and nuanced decision-making in emergency settings. By leveraging these tools, health care providers can improve communication about the potential outcomes of EVT, ultimately leading to better patient care and outcomes.

ARTICLE INFORMATION

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Author Contributions

Dr Mehta wrote the main manuscript. Dr Goel helped write the manuscript and edit. Dr Brown compiled data and helped create figures for the manuscript. Drs Desai and Jadhav edited the manuscript.

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Disclosures

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